

BIG DATA ANALYTICS
[Revised Credit System]
(Effective from the academic year 2021-2022)
SEMESTER - VI

Subject Code	CSE 3272	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS – 03			
Course objectives: This course will enable students to • Understand various components of Hadoop Ecosystem and Spark in the context of solving key issues of Big Data • Implement recommendation system for a real-world problem • Classify Big Data using classification and clustering algorithms. • Analyse graph data efficiently • Demonstrate risk estimation using simulation and analyse data streams			
Module -1			Teaching Hours
INTRODUCTION TO BIG DATA, HADOOP AND SPARK: Introduction to big data, Distributed computing, and Hadoop, Hadoop ecosystem, Hadoop Distributed File System (HDFS), MapReduce framework, MapReduce applications, Understanding YARN architecture, Exploring Hive, Apache Spark background, Uses for Spark, Programming interfaces to Spark, Submission types for Spark programs, Input/Output types for Spark applications Text Book 1: Chapter 1 Text Book 2: Chapter 1			6 Hours
Module -2			
SPARK PROGRAMMING BASICS Anatomy of Spark application, Spark driver, Spark workers and executors, The Spark manager and Cluster manager, Spark applications using the standalone scheduler, Spark applications running on YARN, Deployment modes for Spark applications running on YARN, Introduction to RDDs, Loading data into RDDs, Operations on RDDs. Text Book 2: Chapters 3 and 4			5 Hours
Module – 3			

RECOMMENDATION SYSTEM: The alternating least squares recommender algorithm, Spot checking recommendation, Evaluating recommendation quality, Computing AUC, Hyperparameter selection, Making recommendation Text Book 3: Chapter 3	4 Hours
Module – 4	
PREDICTION WITH DECISION TREES: Fast forward to regression, Vectors and features, Training examples, Decision trees and forests, Preparing the data, Decision tree hyperparameters, Tuning decision trees, Categorical features revisited, Random decision forests, Making predictions. Text Book 3: Chapter 4	4 Hours
Module-5	
ANOMALY DETECTION WITH K-MEANS CLUSTERING: Anomaly detection, K-means clustering, Network intrusion, Choosing K, visualization, Feature normalization, Categorical variables, Using labels with entropy, Clustering in Action Text Book 3: Chapter 5	4 hours
Module -6	
ANALYZING CO-OCCURRENCE NETWORK WITH GRAPHX: Parsing KML documents, Analyzing the MeSH major topics and their co-occurrences, Constructing a co-occurrence network with GraphX, Understanding the structure of networks, Filtering out noisy edges, Small world networks. Text Book 3: Chapter 7	4 Hours
Module -7	
ESTIMATING RISK THROUGH MONTE-CARLO SIMULATION: Methods for calculating VaR: Variance-Co-variance, Historical simulation, Monte-Carlo simulation, Pre-processing, Determining the factor weights, Sampling: The multivariate normal distribution, Visualizing the distribution of returns, Evaluating the results Text Book 3: Chapter 9	4 Hours
Module-8	

STREAM PROCESSING AND MESSAGING USING SPARK: Spark streaming architecture, DStreams, State operations, Sliding window operations, Structured streaming, Apache Kafka Text Book 2: Chapter 7	5 Hours
Course outcomes:	
After studying this course, students will be able: 1. To anticipate the issues that occur while analyzing big data and understand which tools to use based on the problem at hand. 2. To develop a recommendation system for a real-world problem 3. To classify big data by applying algorithms based on supervised and unsupervised techniques. 4. To analyse graph data using GraphX. 5. To apply Monte-Carlo simulation for estimating risk and analyse data streams using spark streaming.	
Text Books: 1. Vignesh Prajapathi, <i>Big Data Analytics with R and Hadoop</i>, Packt Publishing, 2013. 2. Jeffery Aven, <i>Data Analytics with Spark using Python</i>, Pearson, 2018 3. Sandya Ryza, Uri Laserson, Sean Owen and Josh Wills, <i>Advanced Analytics with Spark (2e)</i>, O'Reilly Media Inc, 2017.	
Reference Books: 1. Holden Karau, Andy Konwinski, Patrick Wendell and Matei Zaharia, <i>Learning Spark: Lightning-Fast Big Data Analysis (2e)</i>, O'Reilly Media Inc, 2020. 2. Tom White, <i>Hadoop: The definitive guide (4e)</i>, O'Reilly, 2015.	